Swaminathan Sankaran

DATA SCIENTIST | MACHINE LEARNING ENGINEER | AI ENGINEER
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EDUCATION

University at Buffalo, The State University of New York **Aug 2025 - Dec 2026**
Masters of Science, Engineering Science (Data Science) (GPA: 3.81) *Buffalo, NY*

Vellore Institute of Technology **Aug 2019 - Jul 2023**
Bachelor of Technology, Computer Science and Engineering with Specialization In AI and ML *Chennai, IN*

WORK EXPERIENCE

Zolvit (formerly Vakilsearch) | *Machine Learning Engineer Intern* **Feb 2024 - Aug 2024**

- Eliminated ~23 hours of weekly manual data entry by engineering a fully automated ingestion pipeline for 350+ complex documents, combining Google Vision OCR with a fine-tuned T5-large model to replace a fragile legacy process.
- Drove 99.9% service uptime and eliminated recurring outages by deploying Dockerized ML microservices on AWS EC2 with FastAPI, restart policies, and real-time monitoring via Grafana.
- Automated routing of customer-uploaded documents across categories by achieving 92% classification accuracy, benchmarking Logistic Regression, SVM, and XGBoost on TF-IDF and Doc2Vec features extracted via Google Vision OCR.
- Improved legal precedent retrieval precision by ~28% for lawyers by building a hybrid search engine combining Elasticsearch keyword search with Pinecone vector embeddings across 10,000+ documents.
- Maintained reliability of production ML systems by handling incident response, debugging failures, and improving robustness of legacy pipelines.

PROJECTS

US Electricity Demand Forecast Audit | GitHub **Oct 2026 - Nov 2026**

- Audited 10 years of day-ahead electricity demand forecasts for 50+ US grid operators, analyzing 6.1M hourly time-series records of public U.S. Energy Information Administration (EIA) data in Python and pandas.
- Found 3 errors in EIA's published data, verified against grid operators' own records, that made forecast accuracy appear to decline; once corrected, the 10-year trend was flat.
- Traced California's apparent forecast failure to a 1-hour timestamp shift and battery charging counted as demand (R-squared 0.91); corrected, its 2025 error fell from 8.3% to 2.4%, beating a naive baseline.
- Pre-registered the analysis plan before computing results and built a reproducible pipeline with 138 automated tests (pytest), data-quality rules, statistical trend tests and 8 robustness checks.

Compositional Grounding Probes for Open-Vocabulary Segmentation | GitHub **Aug 2026**

- Built a 914-item benchmark from COCO, LVIS, and Visual Genome to evaluate text-prompted segmentation, with automatically generated labels individually verified by a human reviewer.
- Engineered a reproducible evaluation pipeline with checkpointed GPU batch inference, unit-tested geometry utilities, and automated metrics.
- Exposed systematic model failures on compositional prompts: 20.7% accuracy on negation vs 93% on controls, and 0 of 51 correct on paired spatial queries.
- Traced all failures to the text-grounding stage via an oracle ablation that scored 99.6% with ground-truth inputs.

Drift-Aware MLOps Pipeline | GitHub **Feb 2026**

- Designed a drift-aware MLOps pipeline to detect silent model degradation and trigger automated retraining, improving model reliability under changing data distributions.
- Built and containerized the system using FastAPI, Airflow, MLflow, PostgreSQL, and Evidently AI with Docker Compose, with Kubernetes manifests for production readiness.
- Monitored system health in real time using Prometheus and Grafana dashboards.
- Validated end-to-end on a simulated recession-era covariate and concept shift: retraining lifted shifted-data PR-AUC from 0.84 to 0.89 and ROC-AUC from 0.54 to 0.65, passing the historical-data gate (ROC-AUC 0.75).

CT Scan Tamper Classification | GitHub **Dec 2025**

- Built PyTorch models in a 2-person team to classify 3D lung CT patches as authentic or GAN-tampered (tumors injected or removed), using 169 labeled locations from a 100-scan public dataset.
- Designed a 3D-to-2D compressor that feeds 16-slice CT patches to an ImageNet-pretrained ResNet-18; it ranked first of 5 architectures by peak validation AUC (0.96 on 39 patches from scans held out of training).
- Built the data pipeline end to end: DICOM loading with pydicom, Hounsfield-unit windowing, 3D patch extraction, scan-level train/validation split, class-weighted loss and flip/noise augmentation.

SKILLS

- **Programming Languages**: Python, C/C++, SQL, R
- **Machine Learning & AI**: PyTorch (torchvision, PyTorch Geometric), TensorFlow, Scikit-learn, XGBoost, Hugging Face, embeddings, RAG, Representation Learning (self-supervised & contrastive learning), LangChain, LangGraph, TensorBoard
- **Data Processing & Analysis**: Pandas, NumPy, Medical Imaging (CT, DICOM, pydicom, volumetric data), RDKit
- **Data Visualization & BI**: Matplotlib, Seaborn, Plotly, Streamlit, Tableau, PowerBI
- **MLOps & Cloud**: Docker, Kubernetes, AWS, Apache Airflow, MLflow, Evidently AI, Prometheus, Grafana
- **Developer Tools & Systems**: Linux, Git/GitHub, Pytest, REST APIs, CUDA/GPU training, Coding Agents (Cursor, Claude Code, Codex)
- **Databases & Data Engineering**: PostgreSQL, MySQL, Elasticsearch, Pinecone, Spark, Snowflake

AWARDS & CERTIFICATIONS

- Felix Infausto Scholarship - University at Buffalo, 2025
- Coursera - Probability & Statistics for Machine Learning & Data Science, Machine Learning Specialization, Deep Learning Specialization